So far, I haven't explained the connection between the operator **P** and the classical notion of momentum—namely, mass times velocity, or

$$v = p/m. \tag{9.8}$$

What do we mean by the velocity of a quantum mechanical particle? The simplest answer is that we mean the time derivative of the average position <Ψ|**X**|Ψ>:

$$v = \frac{d\langle\Psi|\mathbf{X}|\Psi\rangle}{dt}$$

or, more concretely, in terms of wave functions,

$$v = \frac{d}{dt}\int \psi^*(x,t)\ x\ \psi(x,t).$$

Why does <Ψ|**X**|Ψ> vary with time? Because ψ depends on time, and in fact we know just how. The time dependence of ψ is governed by the time-dependent Schrödinger equation. We could use that fact to work out how <Ψ|**X**|Ψ> varies with time. I've done it this way—by brute force—and it takes several pages. Fortunately, the abstract methods you learned in earlier Lectures make it easier; in fact, we have already done most of the work in Lecture 4. In fact, before we proceed, I recommend that you review Lecture 4, especially Section 4.9, from the beginning to the appearance of Eq. 4.17. To restate Eq. 4.17,

$$\frac{d}{dt}\langle\mathbf{L}\rangle = \frac{i}{\hbar}\langle\ [\ \mathbf{H},\mathbf{L}\ ]\ \rangle.$$